HI 192 List of Project SS 25-26

<u>DelaCruz/Doria/Rodriguez :</u> (Optimizer and Learning Rate Study of Ovarian Cancer Prediction)

1. STRAMPN-HISTOPATHOLOGICAL IMAGES FOR OVARIAN CANCER PREDICTION

https://ieee-dataport.org/documents/strampn-histopathological-images-ovarian-cancer-prediction

Ovarian Cancer: 481 images
No Ovarian Cancer: 506 images
Total: 987 images
Balanced

75% training, 25% testing; training further split into 70% training 30 % validation

(247 images testing - 740 images training: 518 images training, 222 images validation)

Using transfer learning: **VGG19, EfficientNetB3, ResNet50, DenseNet121**

Fixed: drop-out (0.6, 0.4, 0.3), dense nodes (256, 128, 64), no of epochs:50, input size (256x256); batch size 32

Loading and Augmenting Data (applying Shuffling on Train and Validations Sets only)

```python
train = image_generatorFullAugment.flow_from_directory(
    "./Dataset_split/train",
    batch_size=32,
    shuffle=True,
    class_mode='binary',
    target_size=(256,256)
)

validation = image_generatorFullAugment.flow_from_directory(
    "./Dataset_split/val",
    batch_size=1,
    shuffle=True,
    class_mode='binary',
    target_size=(256,256)
)

test = image_generatorNoAugment.flow_from_directory(
    "./Dataset_split/test",
    batch_size=1,
    shuffle=False,
    class_mode='binary',
    target_size=(256,256)
)
```

Apply pre-processing techniques: normalization, image shuffling, augmentation, train-test split

Should have augmentation on the training images only

In [4]:

```python
from keras.preprocessing.image import ImageDataGenerator

image_generatorFullAugment = ImageDataGenerator(
    rotation_range=30,
    width_shift_range=0.15,
    shear_range=0.3,
    zoom_range=0.30,
    samplewise_center=True,
    horizontal_flip=True,
    samplewise_std_normalization=True
)

image_generatorNoAugment = ImageDataGenerator(
    samplewise_std_normalization=True
)
```

```
/opt/conda/lib/python3.7/site-packages/keras_preprocessing/image/image_data_generator.py:356: UserWarning: This ImageDataGenerato
r specifies `samplewise_std_normalization`, which overrides setting of `samplewise_center`.
  warnings.warn('This ImageDataGenerator specifies '
```

```python
resnet_model = tf.keras.Sequential([
    resnet_base_model,
    GlobalAveragePooling2D(),
    Dense(256, activation="relu"),
    BatchNormalization(),
    Dropout(0.6),
    Dense(128, activation="relu"),
    BatchNormalization(),
    Dropout(0.4),
    Dense(64, activation="relu"),
    BatchNormalization(),
    Dropout(0.3),
    Dense(1, activation="sigmoid")
])

opt = tf.keras.optimizers.Adam(learning_rate=0.0001)
metrics = [
    'accuracy',
    tf.keras.metrics.Precision(name='precision'),
    tf.keras.metrics.Recall(name='recall'),
    tf.keras.metrics.AUC(name='auc')
]
resnet_model.compile(optimizer=opt, loss='binary_crossentropy', metrics=metrics)
```

Main Experiments: to run at LR=0.0001, 0.00001, 0.000001

Adam (Adaptive Moment Estimation optimizer)
Adagrad (Adaptive Gradient optimizer)
Adamax (Maximum adaptive moment estimation optimizer)
AdaDelta
SGD (Stochastic Gradient Descent)
RMSProp (Root Mean Square Propagation optimizer)
TOTAL SIMULATION MODELS: 4 x 3 x 7= 84

Main Metric: Accuracy and AUC; but need to get all metrics (accuracy, recall, precision, specificity, F1-score and AUC)
Binary classification: With Cancer vs Without Cancer

Make sure to generate accuracy and loss curves for every 84 simulation.

Make sure to generate all AUC curves for all 84 simulations

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